Supplementary Material to:
Thermal conductivity of chemical vapor deposition diamond enriched with ^{13}C isotope

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I. OPTICAL ABSORPTION SPECTRA

The optical absorption spectra for sample D13 at 300 K are shown in Fig. 1S. Intensity of a weak band peaked at 270 nm, known to belong to neutral single substitutional nitrogen N_s^0, is on the noise level, and does not exceed 0.1 cm^{-1}, which corresponds to 10^{16} cm^{-3} ($\sim$ 50 ppb) N_s^0. The spectrum is averaged over three independent measurements.

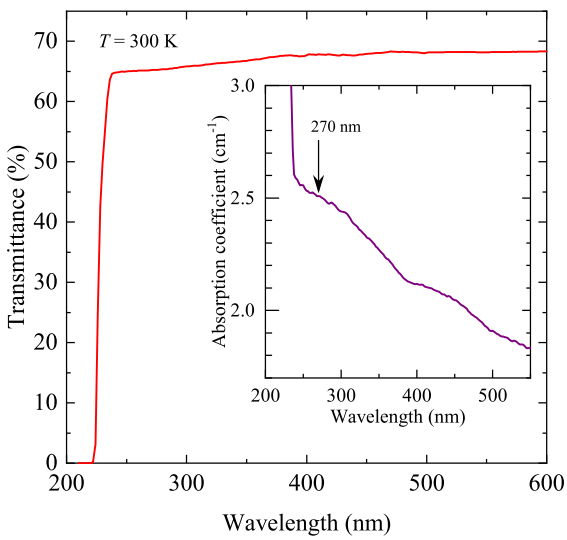

FIG. 1S. Optical transmission and optical absorption (inset) spectra for sample D13.

II. FIRST- AND SECOND-ORDER RAMAN SPECTRA

The first- and second-order Raman spectra for diamond with natural isotopic abundance and for diamond with enriched composition D13 at a temperature of 300 K are shown in Fig. 2S. The isotope shift of the components of the second-order spectrum is in excellent agreement with the experimental data from Vogelgesang *et al.*,[1] see Table 1S.

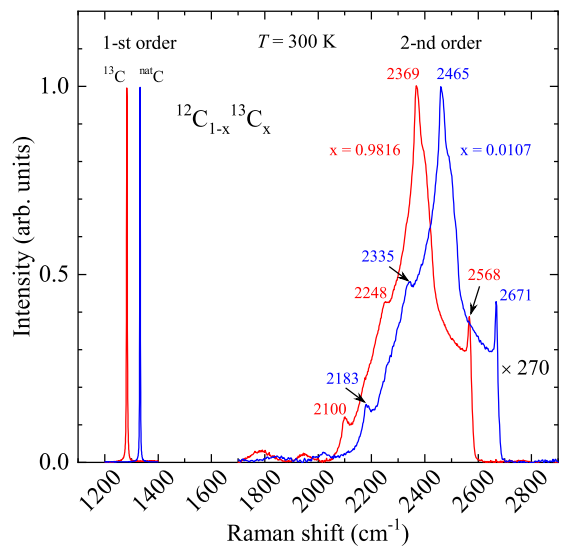

FIG. 2S. Raman spectra for natural isotope composition ^{nat}C (at a concentration of the isotope ^{13}C $x = 0.0107$) and D13 ($x = 0.9816$) diamonds with first-order and weak second-order peaks. The second-order spectrum is enhanced 270 times for better visibility. The spectra are normalized to unity.

TABLE 1S. The features of second-order Raman spectrum as numerated by Vogelgesang *et al.*[1], and their positions ω for natural isotope composition ^{nat}C ($x = 0.0107$) and D13 ($x = 0.9816$) diamonds measured in this work.

Feature	ω_{D13}	ω_{natC}	$\omega_{D13}/\omega_{natC}$
15/16	2100	2183	0.9620
20	2248	2335	0.9627
21	2369	2465	0.9611
25	2568	2671	0.9614

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III. THERMAL CONDUCTIVITY MEASUREMENTS

In Fig. 3S shows an image of sample D13 installed in the experimental cell for measuring thermal conductivity. The sample is pressed against the heat sink using an alligator clip. A gradient heater made of a chip resistor was glued to the free end of the sample using GE7031 varnish. The chip dimensions were 1.50×0.80×0.43 mm^3, the nominal resistance was 1100 Ohm. The thermocouple junctions (J1 and J2) are attached to silver-plated copper wires (70 micron diameter) using GE7031 varnish. The wires are in turn bonded to the wide edge of the specimen using Stycast 2850FT (blue) epoxy; the average distance between thermal contacts was 1.66 mm.

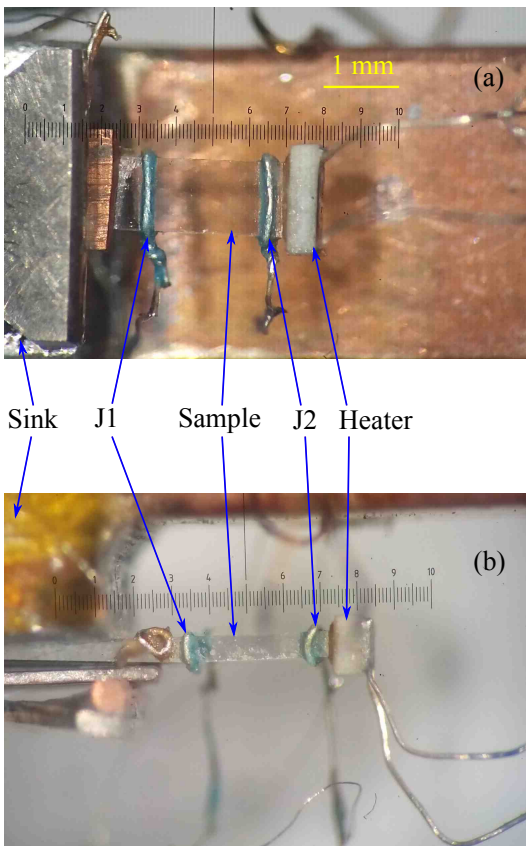

FIG. 3S. Sample D13 installed in an experimental cell for measuring thermal conductivity: (a) – front view and (b) – side view.

IV. MODELING OF $\kappa(T)$

Callaway's expression for the phonon thermal conductivity in its original full form[2] was used to model the measured $\kappa(T)$. The conductivity $\kappa(T)$ is a sum of two terms, the Debye (kinetic) term κ_1 and Ziman (drift) term κ_2:

$$\kappa(T) = \kappa_1 + \kappa_2, \tag{1S}$$

where

$$\kappa_1 = G\,T^3 \langle \tau_C \rangle, \tag{2S}$$

$$\kappa_2 = G\,T^3 \frac{\langle \tau_C/\tau_N \rangle^2}{\langle \tau_C/(\tau_N \tau_R) \rangle}, \tag{3S}$$

$$G = \frac{{k_B}^4}{2\pi^2 \hbar^3 v}. \tag{4S}$$

Here, v is a mean phonon velocity, k_B is the Boltzmann constant. Scattering rates in phonon-phonon processes and other resistive scattering processes are summarized in accordance with the Matthiessen rule. The combined relaxation rate τ_C^{-1} is defined by

$$\tau_C^{-1} = \tau_R^{-1} + \tau_N^{-1}, \tag{5S}$$

where τ_R^{-1} is the total scattering rate of all resistive processes, and τ_N^{-1} is the rate of the three-phonon normal processes. The angled brackets $\langle f \rangle$ represent the following operation:

$$\langle f \rangle = \int_0^{T_D/T} \frac{x^4 e^x}{(e^x - 1)^2} f(x)\, dx, \tag{6S}$$

where $x = \hbar\omega/k_B T$, ω is the phonon frequency.

When fitting the model to the experimental data, the following scattering processes were taken into account:

1. normal and umkplapp three-phonon processes,
2. boundary scattering,
3. point-defect scattering,
4. bound-hole scattering.

A. Three-phonon scattering processes

The scattering rates of normal (τ_N^{-1}) and umkplapp (τ_U^{-1}) three-phonon processes are determined by the following expressions:

$$\tau_N^{-1} = \widetilde{A}_N\, \omega\, T^4 = A_N\, x\, T^5, \tag{7S}$$

$$\tau_U^{-1} = \widetilde{A}_U\, \omega\, T^2 e^{-T_D/\alpha T} = A_U\, x\, T^3 e^{-T_D/\alpha T}, \tag{8S}$$

where $T_D = 2230$ K is the Debye temperature for diamond. The values of parameters $A_N = 3.353 \times 10^{-3}$ s^{-1}K^{-5}, $A_U = 1272$ s^{-1}K^{-3}, and $T_D/\alpha = 560$ K were found to yield the best fitting of the theoretical model to the experimental data on $\kappa(T)$ for highly pure single crystal diamond.[3]

B. Point-defect scattering

To calculate the rate of point-defect scattering from non-isotopic defect $\tau_{\rm pd}^{-1}$ the following expression for g_2 by Turk and Klemens[4] was used

$$g_2 = \sum_i f_i \left(\Delta M_i/M + 2\,\gamma\,\beta_i\right)^2, \tag{9S}$$

where γ is the Grüneisen constant (1.1 for diamond[5]), and β_i is the fractional volume difference between impurity and host atoms. According to Ref. 6, page 1415, $\beta^{(\mathrm{N}_s^0)} \approx 0.35$ for a single-substitutional nitrogen in diamond. With these data, the effect of lattice distortion near impurity nitrogen, the second term in parentheses in Eq. (9S), is 4.6 times greater than the mass difference effect, the first term. This number is very close to the result, about 5, of *ab initio* calculations by Katcho *et al.*[7] for N_s^0. The parameter $a_{\mathrm{pd}}^{(\mathrm{N}_s^0)} = A_{\mathrm{pd}}^{(\mathrm{N}_s^0)}/[\mathrm{N}_s^0]$ determines the rate of point-defect scattering from one nitrogen atom N_s^0 in cm^3, and

$$a_{\mathrm{pd}}^{(\mathrm{N}_s^0)} = 2.77\times10^{-22}\ \mathrm{s}^{-1}\mathrm{K}^{-4}\mathrm{cm}^3. \tag{10S}$$

For a vacancy defect, Ratsifaritana and Klemens[8] showed within perturbation theory that the expression in parentheses of the Eq. (9S) is equal to -3. However, Katcho *et al.*[7] from the first-principal study of vacancy defect in diamond found that this expression is equal to -6. Due to this, the rate of scattering from one vacancy in cm^3 is determined by

$$a_{\mathrm{pd}}^{(\mathrm{V})} = 1.14\times10^{-20}\ \mathrm{s}^{-1}\mathrm{K}^{-4}\mathrm{cm}^3. \tag{11S}$$

This parameter is 41 times higher the corresponding parameter for the scattering from nitrogen atom N_s^0. And a single atom N_s^0 scatters phonons about 125 times more efficiently than a single isotope $^{13}\mathrm{C}$ in one cm^3.

C. Bound-hole scattering

The phonon scattering from charge carriers (electrons and holes) bound to impurity centers[9] is an efficient process that operates in a limited temperature range and has the rate several orders of magnitude higher than rate of scattering by point defects. These scattering processes are called *bound-carrier scattering* or, more specifically, *bound-electron scattering* and *bound-hole scattering*). A broader review of these scattering processes is given in Ref. 10.

In the case of acceptors, the most successful explanation of the mechanism responsible for the suppression of conductivity has been given by Suzuki and Mikoshiba.[11,12] Following Challis and Haseler[13], we used in our calculations the average of expressions for phonon scattering rates for longitudinal and transverse phonon modes given by Suzuki and Mikoshiba.[11] For the elastic (resonant) scattering by holes, the rate is

$$\tau_{\mathrm{bh}}^{-1} = N B_{\mathrm{bh},j}(\omega)\omega^4 F(\omega)\,\frac{W_{\mathrm{bh}}^{(\mathrm{e})}(\omega)}{(\Delta^2-\hbar^2\omega^2)^2}, \tag{12S}$$

where

$$B_{\mathrm{bh},j}(\omega) = \frac{1}{100\pi\rho^2 v_j^2}\,(2D_{u'}^a/3)^4\ f^2(\omega/v_j), \tag{13S}$$

$$f(q_j) = \left(1+(a^*)^2(q_j)^2/4\right)^{-2}, \quad q_j = \omega/v_j, \tag{14S}$$

$$F(\omega) = v_1^{-5}\,f^2(q_1) + \frac{3}{2}v_2^{-5}\,f^2(q_2), \tag{15S}$$

$$W_{\mathrm{bh}}^{(\mathrm{e})}(\omega) = \frac{10}{3}[(3+D^4)\Delta^2 + 6(1+D^2)\hbar^2\omega^2]. \tag{16S}$$

Here ρ is the crystal density (3.5153 $\mathrm{g\,cm}^{-3}$ for diamond); $j = 1,2,3$ stands for the phonon polarization index, v_1 and $v_2 = v_3$ are the average velocities of longitudinal and transverse phonons (18.12×10^5 $\mathrm{cm\,s}^{-1}$ and 12.24×10^5 $\mathrm{cm\,s}^{-1}$ for diamond, respectively); Δ is the energy difference between two doublets of the fourfold degenerate ground state in an undistorted lattice; a^* is the effective Bohr radius of the acceptor in the ground state; N is the number of neutral acceptors per unit volume; $D = D_u^a/D_{u'}^a$; $D_{u'}^a$ and D_u^a are the deformation potentials for T_2-type and E-type lattice distortions, respectively. Assuming $D = 0.86$,[11] we get

$$W_{\mathrm{bh}}^{(\mathrm{e})}(\omega) = 11.8\Delta^2 + 34.8\hbar^2\omega^2. \tag{17S}$$

The Debye velocity $v_D = 13.36\times10^5$ $\mathrm{cm\,s}^{-1}$, and the Casimir velocity $v_C = 12.98\times10^5$ $\mathrm{cm\,s}^{-1}$ for diamond with natural isotopic composition were calculated using the room temperature elastic moduli from Ref. 1. The temperature variation of the elastic moduli and density ρ is so small that the changes of T_D and v_D upon cooling to low temperatures is negligible (about 0.1%). The Casimir velocity v_C is only $\approx 3\%$ lower than v_D, so we used v_D instead of v_C and other average phonon velocities, omitting the subscript in our calculations.

D. Model fitting

In bound-hole model, the total scattering rate τ_R^{-1} of all resistive processes is given by

$$\tau_R^{-1} = \tau_U^{-1} + \tau_b^{-1} + \tau_{\mathrm{pd}}^{-1} + \tau_{\mathrm{bh}}^{-1}. \tag{18S}$$

The parameters l_b, η, A_{pd}, Δ, the product $(D_{u'}^a)^4 N$, and a^* were obtained by fitting the theoretical $\kappa(T)$ calculated by Eq. (1S) to the measured data. The fit was performed using the least squares method, with the above model parameters being the adjustable (or "free") parameters.

The measured thermal conductivity $\kappa(T)$ for samples GPI and D13 is very well approximated by the Callaway model if we add to the consideration the process of phonon scattering by charge carriers bound to impurity doping centers, as well as scattering at sample boundaries, point defects, and in three-phonon processes. Scattering by bound carriers in the context of diamond thermal conductivity is discussed in detail in Refs. 10 and 14.

The adjustable parameters of the best fitting are listed in Table 2S.

E. First-principles calculations for sample D13

First-principles calculations of $\kappa(T)$ for diamond sample D13 were performed using the mean free paths l_b of phonons for diffuse scattering at sample boundaries and the parameter $A_{\rm pd}$ found by fitting the Callaway model to the measured data (Fig. 4S, green curve). This $A_{\rm pd}$ is significantly larger than $A_{\rm iso}$ because the fitting procedure attempted to fit the model to the experimental data near the peak of $\kappa(T)$ by increasing the phonon scattering rate at point defects above the isotopic scattering rate. The calculated dependence $\kappa(T)$ with calculated value of $A_{\rm pd} = A_{\rm iso}$ is shown in Fig. 4S by an orange curve, along with the calculated data within the framework of the Callaway model (dashed curve in the figure).

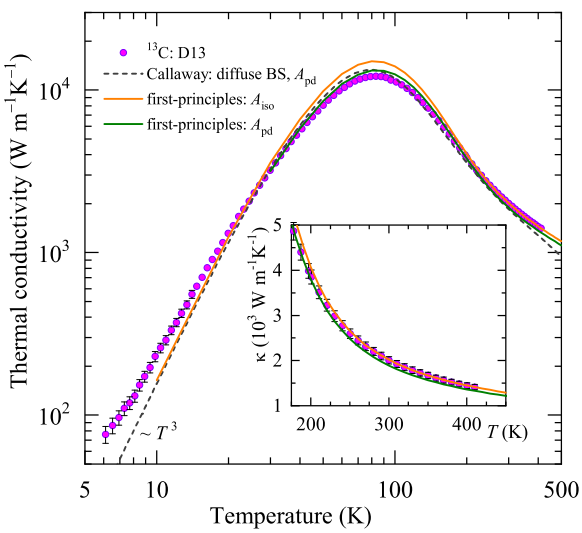

FIG. 4S. Thermal conductivity as a function of temperature for a diamond sample D13 enriched with the ^{13}C isotope. Symbols are measured data, lines are calculation results. The error bars are larger than the symbols size at temperatures below 14 K on the main panel. The inset shows the data on linear scales at high temperatures.

F. ShengBTE calculations: Convergence test

In ShengBTE, $40\times40\times40$ grid was shown to be sufficient to ensure convergence for the thermal conductivity of diamond at temperatures above 100 K.[15] However, to calculate the conductivity with reasonable accuracy (within a few percent, similar to experimental errors in the thermal conductivity measurements) at low temperatures and at the thermal conductivity peak, where boundary scattering has a large effect on the thermal conductivity, a denser grid is needed. Thus, at a temperature of 10 K, sufficient convergence within the error of

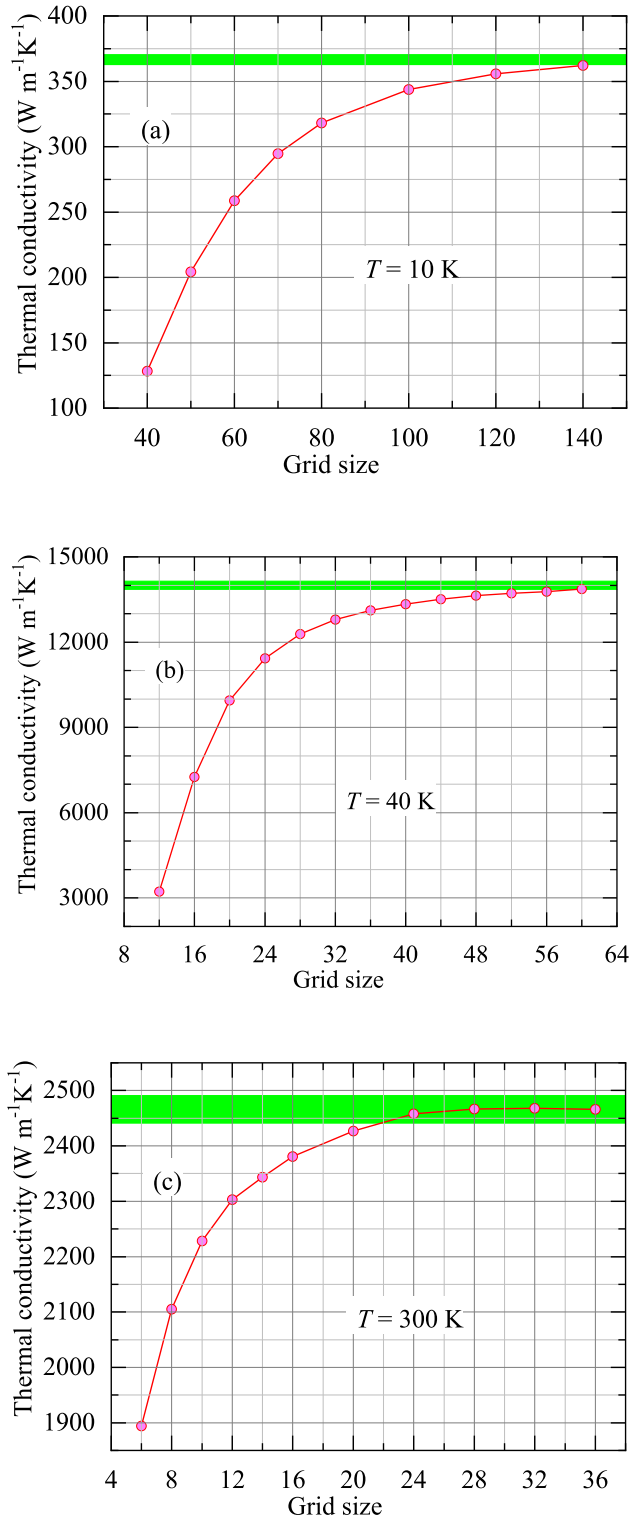

FIG. 5S. Convergence of the calculated thermal conductivity of a GPI diamond sample with natural isotopic composition at 10 K (a), 40 K (b) and 300 K (c) as a function of the q-grid size. The green area denotes the threshold of no more than 1% of the extrapolated ("true") value, which we adopted as the convergence criterion.

TABLE 2S. Fitting parameters of the Callaway model including bound-hole scattering for measured $\kappa(T)$ for the samples GPI and D13. The sd is the standard deviation of the model from measured data. The uncertainties on the last digits of parameters are given in parentheses. The $A_{\rm iso}$ isotope scattering rate parameter calculated from the isotopic composition of samples GPI and D13 is 4.109×10^{-3} s^{-1}K^{-4} and 6.743×10^{-3} s^{-1}K^{-4}, respectively.

Sample	$A_{\rm pd}$ (s^{-1}K^{-4})	$(D^a_{u'})^4N$ (eV^{4}cm^{-3})	Δ (K)	a^* (nm)	l_b (mm)	η (nm)	sd (%)
GPI	$4.37(15)\times10^{-3}$	—	—	—	1.64(5)	4.8(2)	4.2
GPI	4.109×10^{-3}	$4.4(15)\times10^{20}$	300(13)	1.10(6)	2.05(10)	6.0(3)	2.4
D13	$10(5)\times10^{-3}$	—	—	—	0.62(2)	4.4(2)	9.2
D13	$7.4(2)\times10^{-3}$	$4.8(7)\times10^{20}$	300(11)	0.83(3)	0.88(3)	6.6(3)	1.6

about 1% was obtained only for $140\times140\times140$ grid [see Fig. 5S(a)]. At the same time, such a dense grid is more than enough at high temperatures, which is captured by our convergence tests at 40 and 300 K with the error less than 1% [see Figs. 5S(b,c)]. The choice of a 60×60×60 grid allowed us to accurately capture the behavior of thermal conductivity starting from 40 K, near the peak at 70 K, as well as at higher temperatures.

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